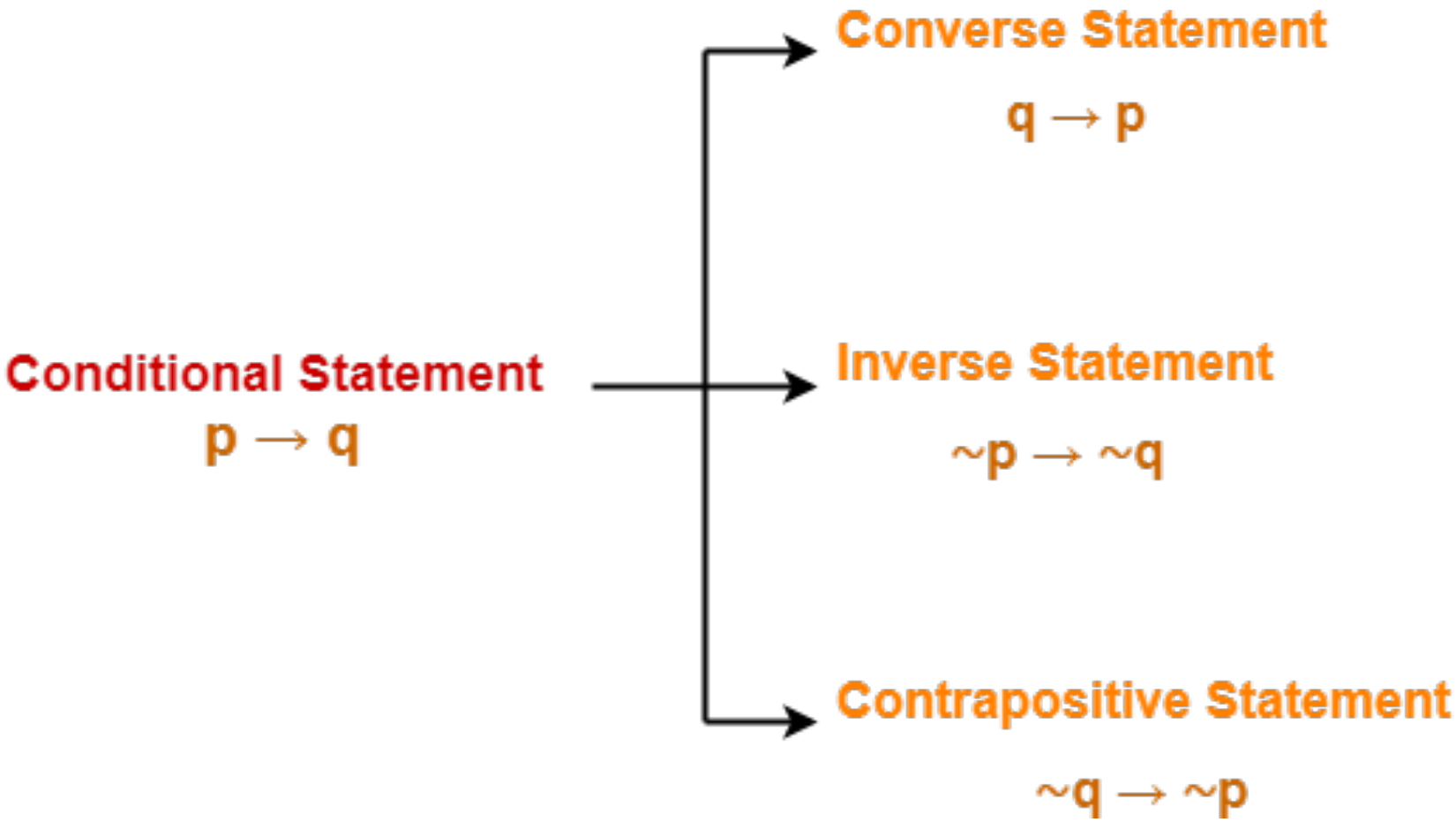
CONVERSE, INVERSE, & CONTRAPOSITIVE of IF-THEN STATEMENTS

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CONVERSE, INVERSE, and CONTRAPOSITIVE of IF-THEN STATEMENTS

CONVERSE: "If p, then q" becomes "if q, then p."
INVERSE: "If p, then q" becomes "if not p, then not q."
CONTRAPOSITIVE: "If p, then q" becomes "if not q, then not p."


Conditional Statement
p → q
Converse Statement
q → p
Inverse Statement
~p → ~q
Contrapositive Statement
~q → ~p

- Given an **if-then statement** "if *p*, then *q*", we can create three related statements:
- A conditional statement consists of two parts, a **hypothesis** or **antecedent** in the "if" clause and a **conclusion** or **consequent** in the "then" clause.

For instance, **"If it rains, then they cancel school."**

"It rains." is the hypothesis.
"They cancel school." is the conclusion.

- To form the **converse** of the conditional statement, interchange the hypothesis and the conclusion.

The converse of

"If it rains, then they cancel school"

is

“If they cancel school, then it rains."

- To form the **inverse** of the conditional statement, take the negation of both the hypothesis and the conclusion.

The inverse of

"If it rains, then they cancel school"

is

"If it does not rain, then they do not cancel school."

- To form the **contrapositive** of the conditional statement, interchange the hypothesis and the conclusion of the inverse statement.

The contrapositive of

"If it rains, then they cancel school"

is

"If they do not cancel school, then it does not rain."

Give the converse, inverse and contrapositive of the given conditional.

Conditional: $p \rightarrow q$ "if p, then q"

"If you are a Filipino, then you are Asian."

Converse: $q \rightarrow p$ "if q, then p"

"If you are Asian, then you are a Filipino."

Inverse: $\sim p \rightarrow \sim q$ "if not p, then not q"

"If you are not a Filipino, then you are not Asian."

Contrapositive: $\sim q \rightarrow \sim p$ "if not q, then not p"

"If you are not Asian, then you are not a Filipino."

Give the converse, inverse and contrapositive of the given conditional.

Conditional:

"If the animal has wings, then it is a bird."

Converse:

"If the animal is a bird, then it has wings."

Inverse:

"If the animal has no wings, then it is not a bird."

Contrapositive:

"If the animal is not a bird, then it does not have wings"

Give the converse, inverse and contrapositive of the given conditional.

Conditional: $p \rightarrow q$

"If the figure is a square, then it has four sides."

Converse: $q \rightarrow p$

"If the figure has four sides, then it is a square."

Inverse: $\sim p \rightarrow \sim q$

"If the figure is not a square, then it doesn't have four sides."

Contrapositive: $\sim q \rightarrow \sim p$

"If the figure doesn't have four sides, then it is not a square"

MORE EXAMPLES…

Find the converse, inverse, and contrapositive of each conditional proposition.

1. If it is Monday, then Julio is working.
2. If a pentagon is regular, then it is equiangular.
3. If it's raining, then the field is wet.
4. Dark chocolate contains 70-80% cocoa.
5. A triangle is a polygon.

1. If it is Monday, then Julio is working.

CONVERSE:

If Julio is working, then it is Monday.

INVERSE:

If it's NOT Monday, then Julio is NOT working.

CONTRAPOSITIVE:

If Julio is NOT working, then it is NOT Monday.

2. If a pentagon is regular, then it is equiangular.

CONVERSE:

If a pentagon is equiangular, then it is regular.

INVERSE:

If a pentagon is NOT regular, then it is NOT equiangular.

CONTRAPOSITIVE:

If a pentagon is NOT equiangular, then it is NOT regular.

3. If it's raining, then the field is wet.

CONVERSE:

If the field is wet, then it's raining.

INVERSE:

If it's NOT raining, then the field is NOT wet.

CONTRAPOSITIVE:

If the field is NOT wet, then it is NOT raining.

4. The equivalent conditional statement is

"If the chocolate is dark, then it contains 70-80% cocoa."

CONVERSE:

If the chocolate contains 70-85% cocoa, then it is dark.

INVERSE:

If the chocolate is not dark, then it does not contain 70-85% cocoa.

CONTRAPOSITIVE:

If the chocolate does not contain 70-85% cocoa, then it is not dark.

5. If it is a triangle, then it is a polygon.

CONVERSE:

If it is a polygon, then it is triangle.

INVERSE:

If it is not a triangle, then it is not a polygon.

CONTRAPOSITIVE:

If it is not a polygon, then it is not a triangle.

❑ **The statement is true, then the contrapositive is also logically true. If converse is true, then the inverse is also logically true.**

